NULL CALIBRATION FOR MULTIPLE-CHOICE EVALUATION: WHEN "ABOVE CHANCE" FAILS THE INPUT-BLIND FLOOR

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ABSTRACT

Multiple-choice scores are commonly interpreted against chance. We argue that this comparison is often insufficient: before an arm is compared with another arm, its score should be tested against the construct's best constant, input-blind predictor. This floor is a necessary, not sufficient, validity condition. Across ten target constructs and one negative control, the floor is not chance and, for content scoring, is under-specified by the tie convention, length unit, and tokenizer; when option count varies, even "chance" is ambiguous. The floor must itself be calibrated: because it is a maximum over k noisy label marginals, it is upward biased even under exactly uniform labels, and only three of the eight letter constructs (MMLU-Pro, MMLU, BoolQ) have floors a balanced null could not produce ($p < 10^{-5}$); on the remaining five the observed floor lies inside the estimator's own sampling noise ($p = 0.14$–0.85), so those constructs establish that chance is the wrong *reference* without establishing that the floor differs from it in magnitude. On MMLU-Pro, all 21 evaluated cells are powered at the scale of the reference effect. Among 15 designated damaged cells in four model families, the null choice alone reverses the reading: damaged arms appear above a chance line while only one clears its best-constant floor, and the honest aggregate is 14/15 at or below the floor. Across five smaller benchmarks, 10/15 designated cells show the same wrong-null flip, while a mandatory power analysis explains why most per-benchmark significance tests are inconclusive. We then adapt a complementary, arm-conditional permutation null for a different question: whether one arm's predictions contain item-level information. Its statistic is the numerator of Cohen's κ evaluated within option-count strata, so it is exactly zero for every pure constant emitter independent of collapse letter; our contribution is the stratification and its use as a pre-comparison gate, not the statistic. We use it to re-judge 27 existing cells. The re-analysis withdraws one above-floor capability label, dissolves both below-floor competence labels, and exposes a weak intact-family anchor. Finally, full-fp32 evaluation removes every bf16 exact tie and changes 18.03% of letter decisions without improving accuracy, ruling out a numerical-tie explanation for the measurement failure. Together, the results motivate a simple reporting rule: calibrate each reported construct to its explicit input-blind null before interpreting arm differences.

1 INTRODUCTION

A recent multilingual evaluation reported an awkward pattern: every tested model was above chance, yet none beat the benchmark's majority-class baseline (Arčon et al., 2026). The authors found the flip because they happened to print both reference lines. This is not merely a curiosity about one newly constructed benchmark. A systematic construct-validity review of 445 language-model benchmarks offers 27 actionable checklist items, but none asks authors to report a null, a chance level, or a constant predictor (Bean et al., 2025). The missing item is operational: *before comparing model arms, test whether the reported construct clears its own input-blind floor*.

For a letter-valued multiple-choice construct, the floor is the most accurate constant letter under the benchmark's empirical gold marginal. For content likelihood, an analogous input-blind heuristic can exploit option length. These references need not equal nominal chance. More importantly, an arm can look "above chance" while carrying no more usable signal through the measured interface than a constant predictor. This distinction is especially consequential under structural damage, where option-token selection bias (Zheng et al., 2024) can dominate the argmax while task-relevant content knowledge remains partly available.

We study null calibration as a *measurement protocol*, not as a new multiple-choice interface or a new debiasing algorithm. Majority-class baselines for MCQA already exist (Balepur et al., 2024); letter-versus-content formulations are established (Gu et al., 2025; Cho et al., 2026); and prior work has shown that constant outputs can game automatic judges (Zheng et al., 2025). Our contribution is to make the construct-appropriate input-blind null a pre-comparison gate, quantify the degrees of freedom hidden inside that null, and separate two questions that require different references:

1. **Is the interface valid for arm comparison?** Use an arm-independent best-constant floor.
2. **Does this particular arm carry item-level information?** Use an arm-conditional permutation null that preserves the arm's prediction marginal.

The distinction prevents two opposite mistakes. Chance can credit a literal constant emitter as competent; conversely, the best-constant floor can label two equally empty emitters differently solely because they collapse onto different letters. The first problem motivates our headline floor. The second motivates read-out v2, for which we prove constant-collapse invariance algebraically.

Our empirical evidence uses ten target constructs plus Winogrande as a negative control, four base-model families, letter and content likelihood interfaces, and structurally damaged arms. The main findings are:

- The null is not "chance." BoolQ's best constant is 0.6217 rather than 0.50; MMLU-Pro's floor is 0.116606, or 1.1661$\times$ naive ten-way chance. We report both percentage-point gaps and ratios because either alone distorts comparisons across option counts. The floor is itself an estimate and needs its own calibration: it is a maximum over k noisy label marginals, hence upward biased even under exactly uniform labels. Calibrated against a balanced null at each construct's own (n, k) (Table 2), MMLU-Pro, MMLU and BoolQ have floors that null could not plausibly produce ($p < 10^{-5}$), whereas the five smaller letter constructs have floors inside the estimator's sampling noise ($p = 0.14$–0.85). We therefore rest the quantitative form of this claim on the first three, and read the other five as showing only that chance is the wrong reference to *report*, not that the floor is far from it.
- Content floors are not dataset constants unless their tie convention, length unit, and tokenizer are fixed. On MMLU-Pro, one tokenizer and one dataset yield floors from 0.125914 to 0.532164 across tie conventions; the latter exceeds the intact model's content score by 32.5 points.
- At MMLU-scale power, 15 designated damaged cells across four families exhibit wrong-null readings: 10/12 non-OLMo cells exceed item-averaged chance (12/12 exceed naive 0.10), yet only 1/12 clears the floor; the three designated OLMo-2 cells exceed either chance line but not the floor. The honest statement is 14/15 at or below the arm-independent floor, with one statistically real but materially negligible exception. Off MMLU, 10/15 designated cells flip and 0/15 clear their floor; the accompanying power table is essential.
- The permutation null makes every pure constant emitter score exactly zero, re-sorts rather than merely shrinks the 27-cell read-out, and reveals that an intact Llama-2-7B is itself a weak anchor on MMLU-Pro.
- Full-fp32 evaluation removes 100% of bf16 exact ties and changes 2,532 letter decisions, but does not recover accuracy. The failure is therefore not repaired by numerical precision.

The floor test is deliberately one-directional. A failure disqualifies the score as evidence for an arm comparison, but a pass does not certify that the benchmark measures the intended capability. Feng et al. (2019) construct a dataset where a partial-input baseline is at chance while artifacts remain exploitable; thus clearing a null is *necessary, not sufficient*.

2 Related Work

Input-blind and constant baselines. Balepur et al. (2024) define a majority-class baseline for letter MCQA and recommend stronger-than-chance references; their object is dataset cheatability under a black-box generative setup, whereas ours is a per-arm gate for likelihood-scored constructs.

The distinction is operational: their invalid-output analysis imputes either 0.0 or 0.25 on MMLU, while the construct's best-constant letter value is 0.2689. Zheng et al. (2025) use the term "null model" for crafted constant responses that adversarially exploit an LLM judge. Their null is an attack optimized against a judge; ours is a reference derived from benchmark statistics and requires no attacker. The metalinguistic preprint that motivates our opening reports the above-chance/below-majority pattern descriptively on one new benchmark, but does not turn it into a pre-comparison protocol (Arčon et al., 2026).

Selection bias and MC interfaces. Letter preference is not a new observation. Zheng et al. (2024) identify selection and token bias and propose PriDe, which estimates an option-ID prior from permuted contents. PriDe fixes the predictor; null calibration fixes the reference line. Even after prediction debiasing, reporting only against chance can leave the benchmark's gold-marginal floor unreported. We therefore do not reject permutation controls on cost grounds: they are a useful and potentially inexpensive diagnostic. They answer whether predictions should be corrected, while our arm-independent floor answers whether the resulting interface score is comparable across arms.

OLMES standardizes both multiple-choice formulation (MCF) and cloze formulation (CF), then uses the better score per task and model (Gu et al., 2025). This is not a size-keyed rule. Its reference discussion is nevertheless framed around a random baseline rather than a measured label-marginal floor, so the max rule can select an interface without testing whether that interface clears an input-blind reference. Our ARC-Easy result shows a case where CF rescues an at-floor letter read-out; MMLU shows that an interface swap need not do so. Cho et al. (2026) independently study symbols, cloze, and hybrid formats and propose a new question-sensitive score. We do not claim the interface contrast; our method instead gates whichever construct is reported.

Length-based scoring. Length-normalized likelihood can over-correct and its token-based form is tokenizer-dependent (Oostermeijer, 2026). We therefore do not claim generic length sensitivity or tokenizer dependence of the scoring rule. We show that the *induced input-blind floor* inherits these choices: its numerical value changes with tie convention, character versus continuation-token length, and tokenizer. OLMES's observation that tokenizer dependence may be irrelevant when ranking options for one fixed model and tokenizer is compatible with our result; our scope is cross-model comparison of the reference itself.

Nulls as interpretive controls. Our arm-conditional read-out v2 is a chance-corrected agreement statistic: Δ_{perm} equals the numerator of Cohen's κ (Cohen, 1960) computed within option-count strata, and the constant-emitter zero we rely on is that literature's standard property rather than a new result. We claim only the stratification by variable option count and the use of the statistic as a pre-comparison gate.

Control tasks and selectivity can reverse conclusions about which representation layer is informative (Hewitt & Liang, 2019). That is the closest structural precedent: their null randomizes supervision for a probe, while ours removes input dependence from an MC construct. Ding et al. (2021) similarly motivate sensitivity and specificity tests for representation similarity. Finally, Bean et al. (2025) provide the construct-validity vocabulary and a broad benchmark checklist. We position null calibration as the missing operational item, not as the invention of construct validity.

3 Null-Calibrated Measurement

Let a benchmark contain items $i = 1, \ldots, n$, gold answers y_i, and a scored interface producing predictions $\hat{y}_i$. Its empirical accuracy is

$$\mathrm{acc} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[\hat{y}_i = y_i]. \tag{1}$$

3.1 READ-OUT V1: THE ARM-INDEPENDENT INTERFACE FLOOR

For a letter construct with legal labels $\mathcal{L}$, define the input-blind constant family $h_L(x) = L$. The construct floor is

$$f_{\text{const}} = \max_{L \in \mathcal{L}} \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[y_i = L]. \tag{2}$$

The v1 statistic is $\Delta_{\text{floor}} = \text{acc} - f_{\text{const}}$. Because f_{const} depends only on the evaluated item set, every arm is compared to the same reference. This arm independence is essential: otherwise arm-to-arm comparisons can change because the null changes with the arm.

For content likelihood, we use an input-blind longest-option family. Its prediction is determined solely by option length, but the family is not fully specified until a tie convention and length unit are fixed; token-based variants additionally require a tokenizer. Section 4 treats these choices as part of the construct definition.

Our decision rule is intentionally asymmetric. An arm that fails to beat the floor cannot support a claim that the interface expresses arm-specific capability above the strongest constant reference. An arm that clears it has passed only a necessary test. It may still exploit other artifacts that no single null enumerates (Feng et al., 2019).

3.2 READ-OUT V2: AN ARM-CONDITIONAL INDEPENDENCE NULL

The v1 floor asks whether an interface is suitable for mutually comparable arm scores. It is not collapse-invariant as a competence statistic: on the full MMLU-Pro item set, always-A, always-E, and always-J contain the same item-level information—none—yet v1 assigns 0.000, -2.111, and -3.807 percentage points respectively because the gold marginals are non-uniform. The separate ten-letter implementation self-test below conditions each letter on items where it is legal.

For the arm-specific question, we preserve the arm's own prediction multiset and remove its alignment with items. MMLU-Pro has variable option count, so permutations are uniform *within* `n_opt` strata s. If n_s is the stratum size and $c^{\text{pred}}_{s,L}$ and $c^{\text{gold}}_{s,L}$ are prediction and gold counts, then

$$\widehat{\text{acc}} = \mathbb{E}[\text{acc} \mid \hat{\boldsymbol{y}} \text{ permuted within } s] = \sum_{s} \frac{1}{n n_s} \sum_{L} c^{\text{pred}}_{s,L} c^{\text{gold}}_{s,L}, \tag{3}$$

and

$$\Delta_{\text{perm}} = \text{acc} - \widehat{\text{acc}}. \tag{4}$$

The statistic is not new; the stratification and its use are. Δ_{perm} is the *numerator* of Cohen's κ evaluated within `n_opt` strata: writing $p_o = \text{acc}$ and $p_e = \widehat{\text{acc}}$, one has $\kappa = (p_o - p_e)/(1 - p_e)$ and hence $\Delta_{\text{perm}} = \kappa\,(1 - p_e)$ identically. We verified the identity on all 27 cells of Table 7; it holds to the printing precision in every row. We therefore claim neither this statistic nor the collapse identity below as new: that chance-corrected agreement vanishes for a constant rater is the textbook property, since $p_o = p_e$ there. Our contribution is (i) stratifying the permutation within variable option count, so letters that are illegal for an item are never credited, and (ii) using the quantity as an *arm-conditional pre-comparison gate* with an explicit materiality bar rather than as an agreement score. We report Δ_{perm} rather than κ to keep it on the same percentage-point scale as the v1 floor, which is what makes the two nulls directly comparable.

Constant-collapse identity. For a pure always-L emitter, $P(\hat{y} = L') = \delta_{L'L}$. Writing $m_{L'}$ for the relevant gold marginal,

$$\widehat{\text{acc}} = \sum_{L'} \delta_{L'L} m_{L'} = m_L = \text{acc}, \qquad \therefore \quad \Delta_{\text{perm}} = 0. \tag{5}$$

This is an algebraic identity for every legal collapse letter, not an empirical regularity, and it is the $\kappa = 0$ property specialised to our stratified estimator. The implementation asserts zero to below 10^{-12} for all ten letters. The loophole "collapse onto a different letter" is therefore closed by construction.

Read-out	Question	Null	Required dependence
v1, headline	Is this interface valid for comparing arms?	Best constant, e.g. always-A 0.116606	Arm-independent
v2, companion	Does this arm's prediction vector contain item-level information?	Prediction vector permuted within n_opt strata	Arm-conditional

Table 1: Two nulls answer two different questions. Values and definitions are from paperC/tcodex_out/EVIDENCE_PACK.md, Section D.

The two nulls must not be conflated. Since $\sum_L p_L m_L \leq \max_L m_L$, the v2 null is always at or below the v1 best-constant floor, with equality only when all prediction mass lies on the best constant. Passing v2 must never be described as clearing the paper's floor.

We estimate uncertainty with 10,000 paired bootstrap resamples, recomputing $\widehat{\text{acc}}$ inside every resample, and verify the decision with 10,000 within-stratum permutations; both use seed 7 and must agree at $\alpha = 0.05$. Significance is not enough at $n = 12032$. We define

$$\texttt{recovery\_fraction} = \frac{\Delta_{\text{perm}}}{\Delta_{\max}}, \tag{6}$$

where $\Delta_{\max}$ is the best alignment achievable by reassigning the observed prediction multiset, and require at least 0.10 times the same-family intact anchor for a material signal. If the intact anchor itself has recovery_fraction< 0.10, relative claims are blocked.

Pre-registration status. The v2 rule, gates, stratification, and materiality constant were committed before re-judging the existing cells and before future unscored milestones. The 27 numbers reported here are nevertheless post-hoc by construction because those cells had already been scored, and the designer had already seen the shape of the collapse defect. We therefore present v2 as a pre-registered rule for future read-outs and a transparent post-hoc re-analysis of the existing cells.

4 Four Under-Specifications of the Reference

The phrase "compare to a baseline" hides choices large enough to reverse a verdict. These choices are part of the measured construct and must be printed with the score.

4.1 Tie convention

For a longest-option heuristic, tied maxima can be split fractionally, resolved by the first or last option, credited whenever gold belongs to the tied set, or counted as wrong. On ARC-Challenge, split, first, last, and wrong place all six OLMo-2 arms significantly above the content floor; credit moves five of six significantly below it. On MMLU-Pro, the same dataset, tokenizer, and length unit yield 0.125914 under wrong and 0.532164 under credit, a 40.6-point span. The credit value is 4.6× the wrong value and exceeds the intact base model's content_norm score, 0.207613, by 32.5 points. A convention that lets an oracle harvest every tie is useful as a reductio, not as a neutral default.

4.2 Length unit

"Longest" may mean characters or continuation tokens. On identical items, split changes by as much as 2.02 points across the studied tasks, while credit changes by 14.53–35.20 points. For OpenBookQA, credit is 0.416 under characters and 0.644 under tokens. The reason is visible in the tie sets: ARC-Challenge has 18.60% tied-longest items under characters and 50.85% under tokens. Thus printing the convention without the unit is not reproducible specification.

The character-unit values were recomputed from raw parquet under len(option text). Counting the continuation's leading space gives the same OpenBookQA value, 0.363500, because the shift is uniform across options. The character and token legs are item-matched by the shared loader and a 12/12 self-test.

Construct	n	$L^\star$	Floor	Chance	Gap (pp)	Ratio	$\mathbb{E}[\hat{f}]$	q_{95}	p
MMLU-Pro, naive	12032	A	0.116606	0.100000	+1.661	1.1661×	0.104457	0.107048	$< 10^{-5}$
MMLU-Pro, item-avg.	12032	A	0.116606	0.110877	+0.573	1.0517×	0.104460	0.107131	$< 10^{-5}$
MMLU	14042	D	0.268908	0.250000	+1.891	1.0756×	0.254355	0.258225	$< 10^{-5}$
OpenBookQA	500	A	0.276000	0.250000	+2.600	1.1040×	0.273191	0.294000	0.383
ARC-Easy	2376	C	0.266414	0.250161	+1.625	1.0650×	0.260611	0.270202	0.140
ARC-Challenge	1172	B	0.265358	0.250156	+1.520	1.0608×	0.265105	0.279010	0.453
CommonsenseQA	1221	B	0.208845	0.200000	+0.885	1.0442×	0.215006	0.226863	0.853
PIQA	1838	B	0.504897	0.500000	+0.490	1.0098×	0.509307	0.522851	0.658
BoolQ	3270	B	0.621713	0.500000	+12.17	1.2434×	0.506981	0.517125	$< 10^{-5}$

Table 2: Construct floors compared with chance, *and with the sampling distribution of the floor estimator itself*. Both percentage-point gaps and ratios are needed across option counts. The last three columns calibrate the estimator. Because $\hat{f} = \max_L \hat{m}_L$ is a maximum over k noisy marginals estimated on a single finite item set, it is upward biased even when the gold labels are exactly uniform, so a positive Gap is not by itself evidence that the floor differs from chance. $\mathbb{E}[\hat{f}]$ and q_{95} are the mean and 95th percentile of $\hat{f}$ under an exactly balanced null at that construct's own (n, k), and $p = \Pr(\hat{f} \geq \text{observed floor})$ under that null (2×10^5 multinomial draws per construct, seed 20260814; `paperC/evidence/floor_winners_curse_calibration.json`). **Only MMLU-Pro, MMLU and BoolQ have floors a balanced null could not plausibly produce** ($p < 10^{-5}$). For the five remaining letter constructs the observed floor lies inside the estimator's own noise ($p = 0.140$–0.853), and on CommonsenseQA and PIQA it is in fact *below* $\mathbb{E}[\hat{f}]$. Those five rows therefore support the weaker claim that chance is the wrong *reference*, while providing no evidence that the floor *differs from* chance in magnitude; the $+0.43$–$+2.60$ pp gaps there are the regime our own evidence pack marks "do not oversell" (`EVIDENCE_PACK.md` §A). Floors and chance values are from `paperC/tcodex_out/EVIDENCE_PACK.md`, Section A.

4.3 Tokenizer

Within the token unit, the floor is tokenizer-valued. On identical items, the `split` floor moves by 0.9003 points in the audited comparison and `credit` moves by up to 10.6 points on four-way tasks and 9.26 points on MMLU-Pro. This is a floor-level consequence of the known tokenizer dependence of length-based scoring (Oostermeijer, 2026). We make no vocabulary-size explanation: the ordering is not monotone. The 32k Llama-2 and 152k Qwen3 tokenizers produce fewer MMLU-Pro longest-option ties than the 100k OLMo-2 and 128k Llama-3 tokenizers.

This degree of freedom creates an asymmetry. A token-based content floor is a property of (dataset, convention, unit, tokenizer). A character-based floor is tokenizer-free under its stated character convention. By contrast, the letter floor is a pure property of the item set's gold labels: it is invariant across all 15 cross-family arms and bit-identical across all 21 MMLU-Pro cells.

4.4 What "chance" means when option count varies

MMLU-Pro is described as up to ten-way, but `n_opt` ranges from three to ten. Naive $1/10$ chance is 0.100000, whereas the item average $\text{mean}(1/\texttt{n\_opt})$ is 0.110877. Against the same always-A floor, 0.116606, the gap is therefore either $+1.661$ points and 1.1661×, or $+0.573$ points and 1.0517×. Both are defensible descriptions of a chance line, so both must be reported. The floor remains a dataset property; its gap to an under-specified chance line does not.

5 Experimental Setup

Benchmarks and constructs. We evaluate letter constructs on MMLU, ARC-Challenge, ARC-Easy, OpenBookQA, CommonsenseQA, PIQA, MMLU-Pro, and Winogrande as a negative control; BoolQ supplies an additional binary label floor. Content-side longest-option constructs are evaluated on MMLU, MMLU-Pro, and the five non-MMLU evidence tasks, with Winogrande retained only as a control. The target inventory comprises ten constructs plus the control; OpenBookQA content is reported under both character and token units rather than counted twice.

Dataset	Unit/tokenizer	`split`	`first`	`last`	`credit`	`wrong`
MMLU-Pro	token, OLMo-2	0.193150	0.195894	0.190824	**0.532164**	0.125914
OpenBookQA	character	0.363500	0.362000	0.362000	0.416000	0.320000
OpenBookQA	token, OLMo-2	0.368000	0.384000	0.360000	**0.644000**	0.228000
ARC-Challenge	character	0.274104	0.272184	0.274744	0.347270	0.220990
ARC-Challenge	token, OLMo-2	0.283902	0.265358	0.296075	**0.543515**	0.142491
Winogrande, control	character	0.491713	0.492502	0.490923	0.581689	0.401736
Winogrande, control	token, OLMo-2	0.501184	0.494081	0.508287	**0.933702**	0.068666

Table 3: The content floor changes with tie convention and length unit. Numbers are from `paperC/tcodex_out/EVIDENCE_PACK.md`, Sections F, H, and the August 14 addendum.

Task	n	`keep8` Δ	CI95 half-width	Detect -1.389 pp?
MMLU	14042	-1.389	1.154	reference
ARC-Easy	2376	-0.800	1.305	yes, borderline
Winogrande, control	1267	$+0.552$	1.184	yes
PIQA	1838	$+2.503$	2.775	no
CommonsenseQA	1221	-1.065	3.399	no
ARC-Challenge	1172	-0.939	3.882	no
OpenBookQA	500	-1.800	6.400	no

Table 4: Mandatory power disclosure for small-benchmark null results. Numbers are from `paperC/tcodex_out/EVIDENCE_PACK.md`, Section F.

Models and structural damage. The four families are OLMo-2-7B, Llama-2-7B, Llama-3-8B, and Qwen3-8B-Base. OLMo-2 arms are prune-then-heal checkpoints, including `keep8`, `keep10`, `keep12`, `keep14`, and `shortgpt16`. Non-OLMo damage is evaluation-time front-N truncation: the intact base model is loaded and its layer list is replaced by the first N blocks, with no fresh block, optimizer, gradient, healing, or damaged checkpoint. This is a regime confound, not a clean family contrast, and every joint table labels it. Qwen3 has 36 layers whereas the other families have 32; consequently `k8` retains 22.2% versus 25.0% of the stack.

Interfaces and scoring protocol. The letter prompt is the content prompt with the labelled option body inserted before `Answer:`; the letter interface scores the candidate continuations " A", " B", and so forth. The content interface omits the labelled body and scores option text. Scoring is likelihood-based throughout: summed token log-probability over each candidate continuation followed by argmax. There is no sampling, no decoding, and no regular expression over generated text.

All results use `chat_template = False`. The evaluated systems are base language models with no supervised fine-tuning and no reinforcement learning from human feedback, so applying a chat template would be an unfair and non-comparable measurement. We also use `add_bos = 0`, `desc_style = none`, fp32 master weights with bf16-autocast forward, and batch size 48. MMLU-Pro cross-family runs use maximum length 2048; every final cell has zero truncation.

Statistics and integrity. For deterministic per-item null predictions we report 10,000-resample paired-bootstrap intervals and two-sided mid-p values with seed 7, plus exact McNemar tests. V2 additionally uses 10,000 within-stratum permutations. Every cell requires shard indices exactly $\{0, \ldots, 7\}$, expected cardinality, zero duplicate IDs, zero NaNs, zero truncations, and `chat_template is False`; partial merges raise an error.

Power design. The five smaller evidence benchmarks are unable to resolve every effect of interest, so null results are never interpreted without the power table. MMLU-Pro supplies $n = 12032$ and makes all 21 letter-floor cells capable of detecting MMLU's -1.389-point reference effect; their 95% half-widths range from 0.083 to 0.968 points.

Designated damaged cells. Throughout, *designated damaged* means an arm that was structurally pruned and is reported as a damaged rung of its family; the designation is fixed by the arm's con-

Family	Arm/regime	Accuracy	Δ vs floor (pp)	CI95 half-width	p	v1 verdict
OLMo-2	`keep12`, heal	0.113115	-0.349	0.765	0.3805	at floor
OLMo-2	`keep10`, heal	0.112450	-0.416	0.727	0.2662	at floor
OLMo-2	`keep8`, heal	0.115442	-0.116	0.582	0.7118	at floor
Llama-2	`k14`, truncation	0.116772	$+0.017$	0.083	0.7072	at floor
Llama-2	`k12`, truncation	0.115858	-0.075	0.083	0.0693	at floor
Llama-2	`k10`, truncation	0.113198	-0.341	0.416	0.1132	at floor
Llama-2	`k8`, truncation	0.107463	-0.914	0.736	0.0168	below floor
Llama-3	`k14--k8`, truncation	0.111868–0.114694	-0.474–-0.191	0.407–0.852	0.1296–0.6631	all at floor
Qwen3	`k14`, truncation	0.118933	$+0.233$	0.191	0.0192	above floor
Qwen3	`k12`, truncation	0.115027	-0.158	0.428	0.4645	at floor
Qwen3	`k10`, truncation	0.118185	$+0.158$	0.233	0.1872	at floor
Qwen3	`k8`, truncation	0.107796	-0.881	0.819	0.0362	below floor

Table 5: MMLU-Pro letter read-out against the bit-identical always-A floor 0.116606. OLMo-2 arms are prune-then-heal; the other families are truncate-only, so regime and family are confounded. Numbers are from `paperC/tcodex_out/EVIDENCE_PACK.md`, Section B.

struction, not by its measured score. Where a denominator such as 14/15 or 10/15 is quoted, it is over exactly this set, and any damaged arm excluded from that denominator is named at the point of exclusion together with its own floor delta, so that no ratio is computed over an undisclosed subset.

Multiplicity. We report per-cell $\alpha = 0.05$ decisions across many cells without a family-wise correction, and the headline counts are therefore counts of per-cell decisions rather than simultaneously valid claims. This is a deliberate and stated limitation: the cells share items, nest arms, and share a null, so the tests are neither independent nor exchangeable, and an off-the-shelf Bonferroni or Benjamini–Hochberg adjustment would not have a defensible family definition here. Readers should treat any individual borderline cell as a per-cell decision. The paper's conclusions rest on the direction and near-unanimity of the aggregate (for instance 0/60 damaged cells clearing their floor), not on any single cell crossing α.

6 Results and Analysis

6.1 Wrong-null verdict flips at MMLU-scale power

MMLU-Pro provides the cleanest test. Under $\mathrm{mean}(1/\mathtt{n\_opt})$, 10/12 damaged non-OLMo cells read above chance; under naive 0.10, 12/12 do. Yet only 1/12 clears the best-constant floor. The three designated damaged OLMo-2 cells are also above either chance line but not above the floor. Across these 15 cells, the reference changes the interpretation; the exact chance-line counts are reported rather than compressed into a universal categorical flip.

The result is strong but not universal. Fourteen of the 15 cells are at or below the floor. The exception, `qwen3_8b_base/k14`, is $+0.233$ points above it ($p = 0.0192$, half-width 0.191). V2 confirms a small alignment effect, $+0.267$ points at $p = 0.0066$, but assigns `recovery_fraction`=0.049, only 9.1% of the intact-family anchor and below the 10% materiality bar. It is a real but immaterial exception, not evidence that `k14` retains MMLU-Pro competence.

The most vivid small-benchmark cases are literal constant emitters. Sixteen damaged cross-family cells have accuracy equal to the marginal of the emitted letter to machine precision. Two OpenBookQA cells emit A on every item, score exactly 0.276000, have $\Delta = 0.000$ points and CI95 $[0, 0]$, and therefore land exactly on the optimal constant. Against chance 0.25, the same empty behavior appears 2.6 points above chance.

Off MMLU, the designated OLMo-2 subset gives the honest count 10/15 above chance and 0/15 above floor. Across the broader non-OLMo replication, 0/60 damaged cells clear their floor while 25/60 read above chance. Only 7/60 are significantly below because 52/60 are underpowered for the MMLU reference effect. This is not an effect-size explanation: ARC-Challenge's median damaged effect is -3.840 points, larger than MMLU's -3.603, but its half-width is 3.92 rather than 1.18 points.

Cell	v1 Δ	v1 p	v1 verdict	Δ_{perm}	v2 p	v2 verdict
Qwen3 `k8`, unhealed	-0.881	0.0362	below floor	-0.139	0.0964	no item-level signal
Llama-2 `k8`, unhealed	-0.914	0.0168	below floor	-0.416	0.1002	no item-level signal
Qwen3 `k14`, unhealed	$+0.233$	0.0192	above floor	$+0.267$	0.0066	trace signal
OLMo-2 `keep14`, healed	$+0.324$	0.3234	at floor	$+0.608$	0.0172	trace signal
Llama-2 intact	$+1.538$	0.0001	above floor	$+1.959$	0.0001	anchor blocked, recovery 0.0545

Table 6: V2 re-sorts rather than uniformly deflates the read-out. All deltas are percentage points. Numbers are from `paperC/tcodex_out/EVIDENCE_PACK.md`, Section D.

6.2 READ-OUT VERSUS RETAINED KNOWLEDGE

A floor-level letter score does not imply that all task-relevant competence is gone. On ARC-Easy, OLMo-2 `keep8` scores 0.2584 on letters, statistically at its 0.266414 letter floor, while `content_norm` reaches 0.6460. The paired gap is $+38.76$ points with McNemar $p = 9.8 \times 10^{-148}$. The arm can rank answer contents but cannot express that knowledge through the damaged letter interface. Conversely, healthy models often favor letters, so "content is the fair interface" is not a general conclusion.

The residual-fraction effect is also construct-specific. On OpenBookQA `content_norm`, using chance rather than the 0.3680 token-longest floor inflates the base arm's residual fraction by $2.11\times$. PIQA and ARC-Easy move in the opposite direction, to $0.90\times$ and $0.98\times$, because their token-longest floors are below chance. Null calibration can increase or decrease the residual; it is not a one-way correction.

6.3 V2 RE-SORTS THE 27-CELL READ-OUT

Under v1, two MMLU-Pro cells are significantly below the best-constant floor. Under v2, the below-null count is zero. `qwen3/k8` moves from -0.881 points ($p = 0.0362$) to -0.139 ($p = 0.0964$), and `llama2/k8` from -0.914 to -0.416 ($p = 0.1002$). Their v1 verdicts remain valid statements about an arm-independent interface floor, but they are not evidence that either arm is worse than its own input-blind prediction marginal. Both competence claims dissolve.

V2 is not uniformly conservative. It withdraws the published above-floor capability label for `qwen3/k14`, replacing it with `TRACE_SIGNAL`, while `olmo2/keep14` moves from at-floor to `TRACE_SIGNAL`. Re-sorting in both directions is evidence that the criterion is doing more than shrinking effects.

It also exposes a benchmark-level limitation invisible to v1. Intact Llama-2-7B has `recovery_fraction`=0.0545 on MMLU-Pro, even though v1 calls it comfortably above floor by $+1.538$ points. Because the intact anchor itself is below 0.10, relative-recovery claims for the Llama-2 family are blocked.

Significance alone would re-import the defect. At $n = 12032$, `qwen3/k14` reaches $p = 0.0066$ at only $+0.267$ points while A wins the likelihood argmax on 94.6% of items. The same data reject simpler diagnostics: the A01 own-modal null over-credits P1 by 1.37 points and flips its sign; for Qwen3 heal@7000 and unhealed `k8`, content is -3.610 and -1.076 points under the same permutation null, compared with letter at -0.022 and -0.139; and `llama3/k12` has modal share 0.339 and normalized entropy 0.614 yet $\Delta_{\text{perm}} = +0.002$ points ($p = 0.997$). Entropy and modal share are descriptive, not competence metrics.

6.4 FULL PRECISION FALSIFIES THE NUMERICAL-TIE MECHANISM

For OLMo-2 `keep8` on MMLU, fp32 removes all 4,303 bf16 exact top-two ties and changes 2,532 of 14,042 letter argmax decisions, or 18.03%. Letter accuracy nevertheless changes by only -0.0015, CI95 $[-0.0064, +0.0033]$, exact McNemar $p = 0.5702$. The arm is more significantly below its floor in fp32: -1.538 points, $p = 0.0060$, versus -1.389, $p = 0.0190$, in bf16. Higher precision can reshuffle ambiguous decisions; it cannot create item-level information that is absent.

6.5 AUDITING THE AUDIT

The analysis uncovered and repaired defects without changing the substantive verdicts. A doubled-tail bootstrap formula counted zero-valued resamples in both tails and produced an illegal $p = 1.042$. Splitting the zero atom evenly makes $p \leq 1$ structural; re-emitting the summaries changed 0/24 verdicts, moved 0/30 p-values across 0.05, and left every non-p field byte-identical.

The first cross-family MMLU-Pro launch also had two integrity failures. A sequence cap validated on the OLMo-2 tokenizer silently left-truncated 10/15 cells; raising it to 2048 and making `n_trunc` a hard per-shard assertion changed 0/14 verdicts and at most 0.0083 points, but benignity was unknowable in advance. Separately, intact Llama-2 exhausted memory on 5/8 shards because it lacks grouped-query attention; the merge guard correctly refused the 3/8 partial result. Setting `use_cache=False` recovered the cell. All reported cross-family MMLU-Pro results use the corrected launch.

7 DISCUSSION

A reporting rule, not a universal theory of failure. The practical recommendation is small: define the strongest construct-appropriate constant or input-blind predictor, report its value and specification, test the arm against it, and only then compare arms. The rule does not say every damaged model becomes constant, every MC interface is invalid, or every chance line inflates a claim. It says that an arm score without its appropriate floor can support the wrong verdict.

The letter-floor and permutation read-outs partition scope. V1 must be arm-independent because it certifies mutual comparability of an interface. V2 must be arm-conditional because it asks whether one prediction vector aligns with its items beyond its own output marginal. Substituting one for the other creates the defects we observe: chance credits constants, while a best-constant competence test confounds knowledge with collapse-letter identity.

Relationship to prediction debiasing. PriDe-style option permutations estimate and remove a model's selection bias (Zheng et al., 2024). This is complementary, not redundant. A corrected predictor can still be evaluated against the wrong reference; an explicit floor remains necessary. Conversely, a floor test does not repair a biased predictor. A complete evaluation can report both predictor-side debiasing and reference-side null calibration.

Power is part of construct validity. Small benchmarks produce apparently reassuring at-floor results even when observed effects are large. Any interpretation of a null result must include the achieved interval. MMLU-Pro changes the evidential status: OLMo-2 `keep8`'s interval excludes an MMLU-sized below-floor effect, so this is a powered non-replication, not an absence of evidence. The broader ladder remains a cliff rather than a gradient; no depth curve should be fitted to the sparse rungs.

Current claim versus retracted claims. The flat ledger in Appendix A.4 records the self-falsification history. The surviving claim is narrower and stronger: under structural damage, the letter interface frequently fails its own arm-independent best-constant floor while remaining above a conventional chance line. This is 14/15 rather than universal at MMLU-Pro, is not equivalent to literal constant emission, and does not by itself identify healing or family as a cause.

A healed non-OLMo arm is in training, but it is not part of this paper's evidence. Its remaining question is only whether it develops material item-level signal. The original `H_heal`/`H_family` dichotomy is unavailable because its unhealed comparator does not fall below the collapse-proof permutation null.

8 LIMITATIONS

Character-unit reconstruction. The OpenBookQA character floor was added after the original audit identified an evidence hole. It is now recomputed from raw parquet and passes a 12/12 character/token self-test, so we do not use the earlier "unrecomputable" status. The remaining limitation is narrower: the character values are reconstructed from raw option text rather than retained as per-item fields in the scored records. Item matching to the token leg therefore follows the shared loader and self-test rather than a stored character-count column. The convention of excluding the continuation's leading space was explicit; including it yielded the identical 0.363500 value.

One family at one depth cannot identify healing. The prospective healed contrast contains one non-OLMo family at one absolute depth. Llama-2, Llama-3, and the `k10`/`k12`/`k14` rungs remain confounded. The observed ladder is a cliff, not a gradient, and we fit no depth curve.

The healing corpora are unmatched and cannot be matched from available data. The Qwen3 arm heals on 5.72 epochs of 5.541B SlimPajama tokens, whereas OLMo-2 used 1.0 epoch of 31.7B Dolmino tokens. Qwen3 cannot consume OLMo-2-token Dolmino and raw Dolmino text is on neither disk. Relative depth is also untested: Qwen3 has 36 layers and OLMo-2 has 32, so `keep8` retains 22.2% versus 25.0%.

The planned causal contrast has dissolved. `H_heal` is not merely unmeasured. Its antecedent required the unhealed Qwen3 `k8` comparator to sit below an arm-conditional null; it does not (-0.139 points, $p = 0.0964$). No amount of further training repairs a comparator that fails the criterion. `H_family` is equally unavailable because 0/27 cells are significantly below the permutation null. A future read-out can ask only whether the healed arm has material item-level signal.

Regime, pooling, and literature scope. Non-OLMo cross-family arms are truncate-only, whereas OLMo-2 arms are prune-then-heal; tables that include both cannot identify a family effect. Pooled results average per-item deviations over disjoint benchmarks and must not be read as per-benchmark verdicts. Finally, the Cho et al. overlap audit used arXiv v4 dated January 12, 2026; the January 26, 2026 camera-ready could not be diffed because the OpenReview PDF endpoint was inaccessible from the audit network.

9 Ethics Statement

This work studies evaluation methodology and uses existing benchmark records and base-model checkpoints. It introduces no human-subject data collection and makes no deployment claim about the evaluated models. The main ethical risk is epistemic: a poorly calibrated reference can make a damaged or degenerate system appear capable, which may distort model comparisons and downstream decisions. We mitigate that risk by publishing the null definition, uncertainty, power, integrity checks, counterexamples, and a flat retraction ledger. We also avoid treating a floor pass as certification, since untested artifacts may remain.

The experiments consumed substantial accelerator resources in their original execution. The manuscript itself adds no new training or scoring result. Our recommended nulls are primarily computed from existing labels, option metadata, and prediction records, so they can usually be added with negligible compute relative to model evaluation.

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A APPENDIX

A.1 COMPLETE READ-OUT V2 TABLE

Table 7 reports all 27 cells used in the post-hoc v2 re-analysis. G1 and G2 fire on no cell; all 27 pass the integrity gate; the permutation and bootstrap decisions disagree on 0/27 cells at $\alpha = 0.05$. The largest absolute difference between their p-values is 0.0371, away from the decision boundary. Stratification is nevertheless load-bearing: for unhealed Qwen3 `k8`, the stratified statistic differs from the unstratified one by -1.271 points because E is not legal on every item.

Family	Cell	Acc.	$\widehat{\mathrm{acc}}$	Δ_{perm}	Half-width	p	Verdict
Qwen3	heal@5000	0.115276	0.115993	-0.072	0.231	0.5412	no signal
Qwen3	heal@5500	0.115691	0.115654	$+0.004$	0.316	0.9810	no signal
Qwen3	heal@6000	0.114860	0.115857	-0.100	0.281	0.4932	no signal
Qwen3	heal@6500	0.114943	0.115864	-0.092	0.287	0.5204	no signal
Qwen3	heal@7000	0.115775	0.115995	-0.022	0.241	0.8522	no signal
OLMo-2	`keep8`@45000	0.117271	0.114046	$+0.322$	0.370	0.0856	no signal
OLMo-2	`keep8`@121000	0.115442	0.113152	$+0.229$	0.446	0.3122	no signal
Qwen3	`k8`, unhealed	0.107796	0.109185	-0.139	0.162	0.0964	no signal
Qwen3	intact	0.461104	0.112212	$+34.889$	0.844	0.0001	item-level signal
OLMo-2	intact	0.271858	0.112324	$+15.953$	0.761	0.0001	item-level signal
OLMo-2	`keep10`@83500	0.112450	0.111508	$+0.094$	0.481	0.6966	no signal
OLMo-2	`keep12`@124000	0.113115	0.112219	$+0.090$	0.471	0.7026	no signal
OLMo-2	`keep14`@200000	0.119847	0.113766	$+0.608$	0.500	0.0172	trace signal
OLMo-2	`shortgpt16`@200000	0.153341	0.112797	$+4.054$	0.575	0.0001	item-level signal
Qwen3	`k10`, unhealed	0.118185	0.116155	$+0.203$	0.228	0.0804	no signal
Qwen3	`k12`, unhealed	0.115027	0.114880	$+0.015$	0.379	0.9432	no signal
Qwen3	`k14`, unhealed	0.118933	0.116263	$+0.267$	0.188	0.0066	trace signal
Llama-2	intact	0.131981	0.112392	$+1.959$	0.525	0.0001	anchor blocked
Llama-2	`k8`, unhealed	0.107463	0.111620	-0.416	0.495	0.1002	no signal
Llama-2	`k10`, unhealed	0.113198	0.114575	-0.138	0.356	0.4510	no signal
Llama-2	`k12`, unhealed	0.115858	0.116528	-0.067	0.081	0.1052	no signal
Llama-2	`k14`, unhealed	0.116772	0.116408	$+0.036$	0.085	0.3950	no signal
Llama-3	intact	0.329205	0.112014	$+21.719$	0.813	0.0001	item-level signal
Llama-3	`k8`, unhealed	0.113531	0.115399	-0.187	0.351	0.2894	no signal
Llama-3	`k10`, unhealed	0.111951	0.112269	-0.032	0.366	0.8746	no signal
Llama-3	`k12`, unhealed	0.111868	0.111851	$+0.002$	0.514	0.9968	no signal
Llama-3	`k14`, unhealed	0.114694	0.110945	$+0.375$	0.446	0.0970	no signal

Table 7: Complete 27-cell read-out v2. Deltas and half-widths are percentage points. “Anchor blocked” denotes absolute item-level signal but no admissible relative-recovery claim. Numbers are from `paperC/tcodex_out/EVIDENCE_PACK.md`, Section D.

A.2 REJECTED V2 ALTERNATIVES

Alternative	Cell	Null/statistic (pp)	Comparator (pp)	Measured reason for rejection
Own-modal null	Qwen3 `k8`	$+1.230$ vs always-E	v2 -0.139	Over-credits by 1.37 pp and flips the sign
Content interface	Qwen3 heal@7000	content -3.610	letter -0.022	More negative under the same permutation null
Content interface	Qwen3 `k8`	content -1.076	letter -0.139	More negative under the same permutation null
Content interface	OLMo-2 `keep8`@121000	content -0.086	letter $+0.229$	Does not rescue the damaged readout
Entropy/modal share	Llama-3 `k12`	$\Delta_{\text{perm}} = +0.002$	$p = 0.997$	Modal share 0.339 and entropy 0.614 still contain no signal

Table 8: Measured reasons for rejecting simpler v2 alternatives. Numbers are from `paperC/tcodex_out/EVIDENCE_PACK.md`, Section D.

A.3 Integrity repairs

Defect	Consequence	Repair/hardening	Measured impact
Bootstrap zero atom counted in both tails	Illegal $p = 1.042$	Two-sided mid-p splits zero mass evenly; $p \leq 1$ structurally	0/24 verdicts changed; 0/30 crossed 0.05; non-p fields byte-identical
Tokenizer-specific left truncation	10/15 first-launch cells lost part of labelled option body	Raise cap to 2048; `n_trunc` becomes a hard per-shard assertion	0/14 verdicts changed; maximum 0.0083 pp; all nine flips on affected items
Llama-2 KV-cache OOM	5/8 shards failed; partial 3/8 result unavailable	Merge guard refused output; `use_cache=False` for teacher-forced pass	Peak memory 94 GiB to 41–50 GiB; intact cell recovered

Table 9: Integrity defects and repairs. Numbers are from `paperC/tcodex_out/EVIDENCE_PACK.md`, Sections E and I.

A.4 What we claim and what we retract

#	Claim previously considered	Current status	Manuscript-safe form
1	Letter behavior is a family-general step function or sharp phase transition.	Retracted	Report per-family descriptions only.
2	Damage turns the letter interface into a constant predictor.	Narrowed	Damage drives the letter score to or below its best-constant floor; modal share is separate.
3	Exact ties are the family-general causal mechanism.	Retracted	Precision changes ties and argmaxes but not accuracy; mechanisms are family-specific.
4	Letter MC is generally an unreliable instrument.	Retracted	Intact models can favor letter or content; calibrate each reported construct.
5	Accuracy-versus-normalized-accuracy length sensitivity is our finding.	Preempted	Attribute generic over-correction and tokenizer dependence to Oostermeijer; our result is floor inheritance.
6	Content is within ± 3 pp of letter on every damaged arm.	Re-scoped	False off MMLU; ARC-Easy reaches $+38.76$ pp.
7	Damaged letter scores are significantly below floor across small non-MMLU benchmarks.	Retracted	Only ARC-Easy `keep10` in OLMo-2; accompany other nulls with power.
8	`k14` is the last arm above floor in every family.	Retracted	Descriptive only for healed OLMo-2; other ladders are cliff-like and regime-confounded.
9	A pooled floor verdict can be read per benchmark.	Prohibited	Pool only per-item deviations and label the aggregate scope.
10	A content floor is a dataset property.	Narrowed	Token floors require dataset, convention, unit, and tokenizer; letter floors are dataset properties.
11	Larger BPE vocabularies create more option-length ties.	Retracted	Tokenizer dependence stands; the vocabulary-size mechanism is non-monotone.
12	Significant below-floor behavior is MMLU-specific.	Retracted and reinterpreted	MMLU-Pro has v1 below-floor cells, but v2 shows no negative item-level signal.
13	Damage universally drives letter to or below floor.	Narrowed	The honest MMLU-Pro count is 14/15; Qwen3 `k14` is a real but immaterial exception.
14	First-launch cross-family MMLU-Pro numbers are usable.	Retracted	Use only the 2048-token corrected launch and canonical v2 evidence.
15	Clearing the floor certifies validity.	Retracted	Clearing is necessary, not sufficient; failure disqualifies, success does not certify.
16	Deprecated numeric claims remain usable.	Prohibited	Do not resurrect $4.8\times$, 0.2822, 58/91, MMLU 0.25, BoolQ 0.50, or 45.74 pp.

Table 10: Flat current-status ledger. The 16 rows and their status are from `paperC/tcodex_out/EVIDENCE_PACK.md`, Section K.

A.5 Reporting checklist

For each reported multiple-choice construct, we recommend the following minimal record:

1. Define the input-blind predictor family and report the best member on the evaluated item set.
2. Report the score, null value, paired effect, confidence interval, and test.
3. For content length nulls, print tie convention, length unit, and tokenizer. If option count varies, print the definition of chance.
4. Establish that the experiment could resolve the effect under discussion; otherwise describe the result as a non-observation.
5. Keep arm-independent interface validity separate from arm-conditional item-level information.
6. Treat a floor pass as necessary, not sufficient.

A.6 Evidence provenance

The construct floors, MMLU-Pro cell table, read-out v2 values, fp32 comparison, small-benchmark power analysis, integrity repairs, and claim ledger are transcribed from the machine-extracted `paperC/tcodex_out/EVIDENCE_PACK.md`. The character-versus-token table additionally uses the pack's August 14 addendum, which records the raw-parquet reconstruction and 12/12 self-test. No healed Qwen3 step-121000 result is included.